

Reference excerpt 02-016

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time instants relating to each event were denoted t_{Pi} , t_{Ri} and t_{Ti} , where i indicates the i -th cardiac cycle between those traces. JVP quantitative evaluation The waves a , c , x , v and y and the intervals a and v are the parameters that characterize the JVP (see Fig. 2). In this paper, these parameters are denoted in terms of the value s assumed by the CSA during the cardiac cycle. The values of the CSA of the IJV in correspondence of the waves a , c , x , v and y , during the i -th cycle are indicated by the parameters a_i , c_i , x_i , v_i and y_i . These parameters are detected on the JVP trace by means of an algorithm based on the following definitions: $a_1 = \max(CSA(t)) : 0 < t < T$ $c_i = \max(CSA(t)) : t_{yi-1} + t_{ya} - T_c < t < t_{yi-1} + t_{ya} + T_c$ $c_i = \max(CSA(t)) : t_{ai} + t_{ac} - T_c < t < t_{ai} + t_{ac} + T_c$ $x_i = \min(CSA(t)) : t_{ai} + t_{ax} - T_c < t < t_{ai} + t_{ax} + T_c$ $v_i = \max(CSA(t)) : t_{xi} + t_{xv} - T_c < t < t_{xi} + t_{cx} + T_c$ $y_i = \min(CSA(t)) : t_{vi} + t_{vy} - T_c < t < t_{vi} + t_{vy} + T_c$ (1) Figure 1: The figure represents an illustrative sequence of three sonograms from a video-clip obtained by a transverse ultrasound scan of the IJV. In each picture it is